LEARNER ASSESSMENT SUBMISSION

KM01: Introduction to Data Science and Data Analysis | Formative Assessment

Unit Standard	251102-001-00-KM-01
NQF Level / Credits	Level 4 / 6 Credits
Assessment Type	Formative — Knowledge Module
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SDP Name	AjanaNova Skills Development Provider
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Total Marks Available	30

Learner Declaration: I declare that this is my own work. I have not copied from any other learner or source without proper acknowledgement. I understand that submitting work that is not my own constitutes academic dishonesty and may result in disqualification.

Signed: *S. Dlamini* **Date:** 14 April 2026

Question 1 (10 marks)

Define data science and explain how it enables evidence-based decision-making in organisations. Your answer should include at least one real-world business application.

Data science is an interdisciplinary field that combines statistics, computer programming, and domain-specific knowledge to extract meaningful insights and patterns from large sets of data. It brings together methods from mathematics, data engineering, and subject-matter expertise to turn raw information into actionable knowledge.

In an organisation, data science enables evidence-based decision-making by replacing guesswork and intuition with measurable, data-driven evidence. Instead of managers relying on experience alone, data science allows them to analyse historical trends, identify patterns, and make predictions about future outcomes. This reduces risk and increases the accuracy of strategic decisions.

One analytical technique commonly used is regression analysis, which helps organisations forecast future values based on historical data. Data visualisation tools such as dashboards and graphs also allow decision-makers to quickly interpret complex information.

A real-world example is a retail company like Shoprite using data science to predict which products will sell more in specific regions during certain seasons. By analysing point-of-sale data, customer purchase history, and supply chain information, the company can make better stock management decisions, reduce waste, and improve customer satisfaction.

Marks awarded: / 10

Question 2 (10 marks)

List and briefly describe the five stages of the data analysis lifecycle. For each stage, explain what happens and why it matters.

The data analysis lifecycle consists of several important stages that guide an analyst from raw data to useful conclusions. Below are the stages I have identified:

Stage 1 — Data Collection: This is when the analyst gathers all the raw data needed to answer the business question. Data can come from databases, surveys, online sources, or company records. Without good data collection, the rest of the process will produce unreliable results.

Stage 2 — Data Cleaning: Raw data is almost never perfect. In this stage, the analyst removes duplicate records, fixes errors, fills in missing values, and converts data into a consistent format. This stage is critical because "garbage in, garbage out" — poor data leads to wrong conclusions.

Stage 3 — Data Analysis and Modelling: This is the core analytical stage where statistical techniques and models are applied to the cleaned data to find patterns, trends, and answers. The analyst may look at the data through summary statistics and charts to understand what it looks like, and then build predictive or descriptive models to draw conclusions.

Stage 4 — Interpretation of Results: After analysis, the analyst interprets what the numbers and models mean in the context of the original business question. This involves critically evaluating whether the findings make sense and whether any assumptions were violated during modelling.

These four stages, when followed in sequence, ensure that data analysis is systematic, reliable, and meaningful for the organisation.

Marks awarded: / 10

Question 3 (10 marks)

Distinguish between structured and unstructured data. Provide two examples of each. Explain which type is more challenging to analyse and justify your answer.

Structured Data: Structured data is information that is organised in a clearly defined format, usually arranged in rows and columns with a fixed schema. Each field has a specific data type and meaning, making it easy for computers and analysts to search, sort, and query.

Examples of structured data:

- A Microsoft Excel spreadsheet containing employee salary information, where each row is an employee and each column is a field such as name, department, and salary.
- A JSON file from a company's API that returns customer order records with defined fields like order_id, customer_name, product, and amount.

Unstructured Data: Unstructured data does not follow a predefined format or schema. It can take many different forms and is much harder for computers to process automatically without special tools.

Examples of unstructured data:

- Customer complaint emails sent to a company's support inbox, which contain free-form text in varying lengths and formats.
- Video recordings from a company's training sessions, which contain visual and audio information with no consistent structure.

Which is more challenging to analyse: Unstructured data is more difficult to analyse because it comes in many different formats and cannot be processed using standard spreadsheet or database tools. It is harder to work with and generally requires more effort and specialised software to make sense of.

Marks awarded: / 10

Assessor Sign-off

Assessor Name & Signature	Date of Assessment	Overall Result (C / NYC)